

BÁO CÁO LAB 8 — ASP.NET CORE RAZOR PAGES & MVC

THÔNG TIN BÀI LAB

Tên bài lab

Lab 8 — ASP.NET Core Code First

Mục tiêu

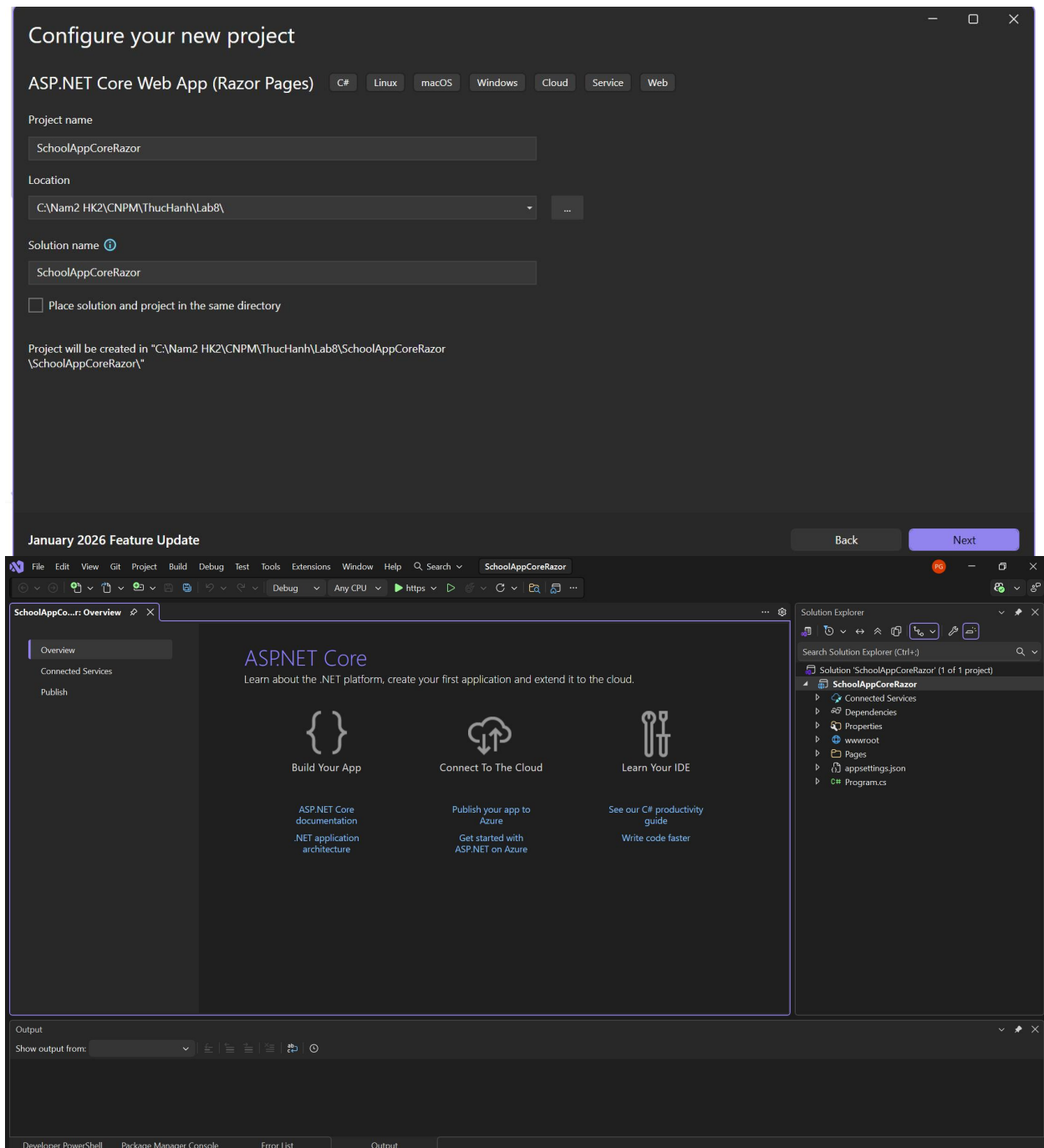
- Tìm hiểu ASP.NET Core Razor Pages và MVC.
- Sử dụng Entity Framework Core theo phương pháp Code First.
- Kết nối SQL Server.
- Tạo CRUD tự động bằng Scaffold.
- Quản lý dữ liệu bằng Razor Pages và MVC.

EXERCISE 1 — ASP.NET CORE RAZOR PAGES

BƯỚC 1 — Tạo Project Razor Pages

Nội dung

Đầu tiên, em tạo project ASP.NET Core Razor Pages trong Visual Studio với tên:



BƯỚC 2 — Cài đặt Entity Framework Core

Nội dung

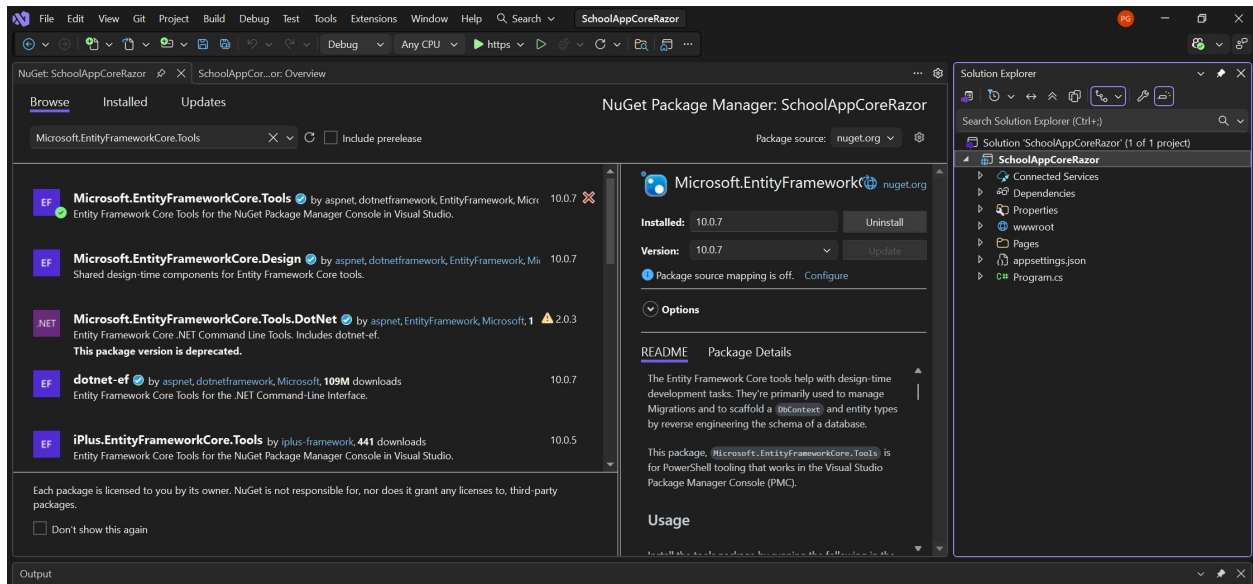
Sau khi tạo project, em cài đặt các package cần thiết bằng NuGet Package Manager:

- Microsoft.EntityFrameworkCore.SqlServer
- Microsoft.EntityFrameworkCore.Tools

- Microsoft.VisualStudio.Web.CodeGeneration.Design

Các package này hỗ trợ:

- kết nối SQL Server,
- migration database,
- scaffold CRUD tự động.



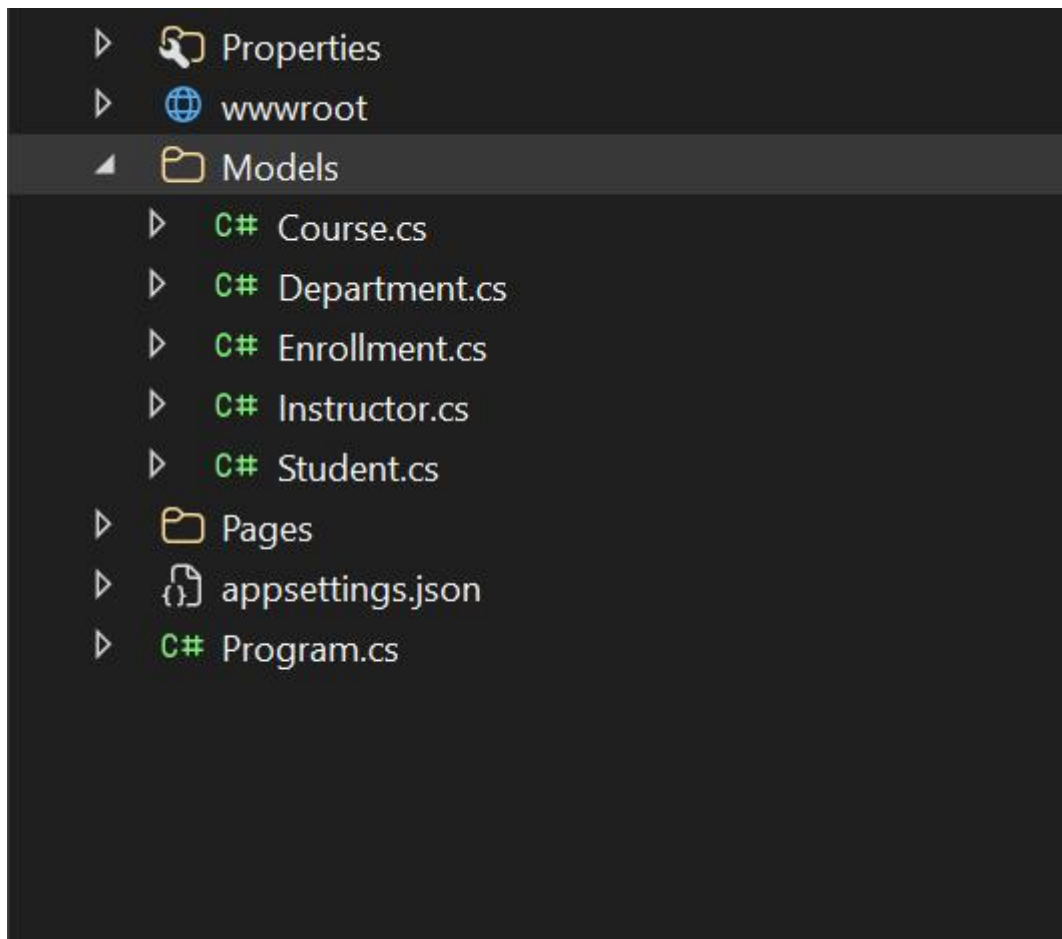
BƯỚC 3 — Tạo Models

Nội dung

Tiếp theo, em tạo folder Models và thêm các class:

- Department.cs
- Course.cs
- Student.cs
- Instructor.cs
- Enrollment.cs

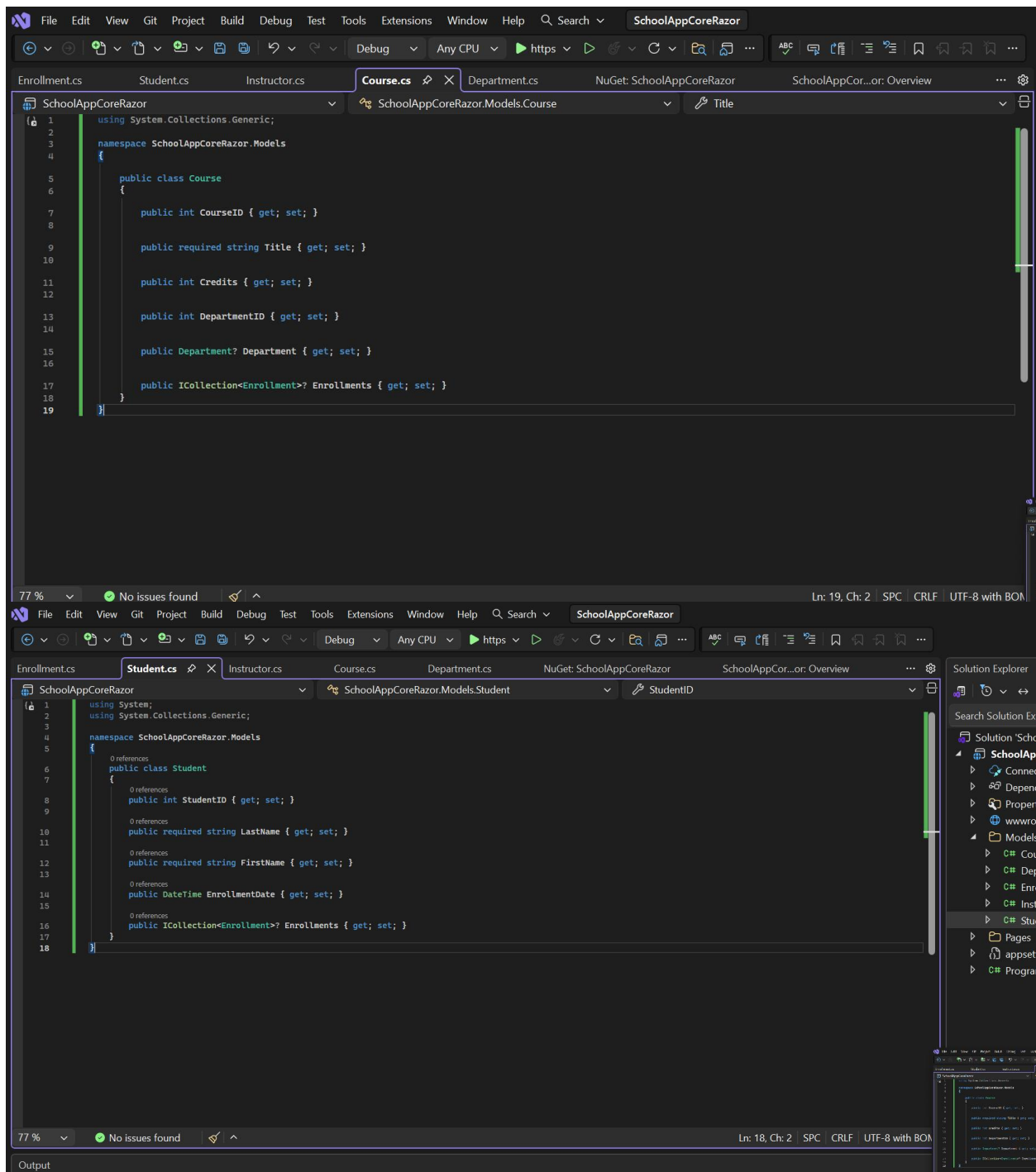
Các class này dùng để mô tả dữ liệu trong hệ thống quản lý trường học.

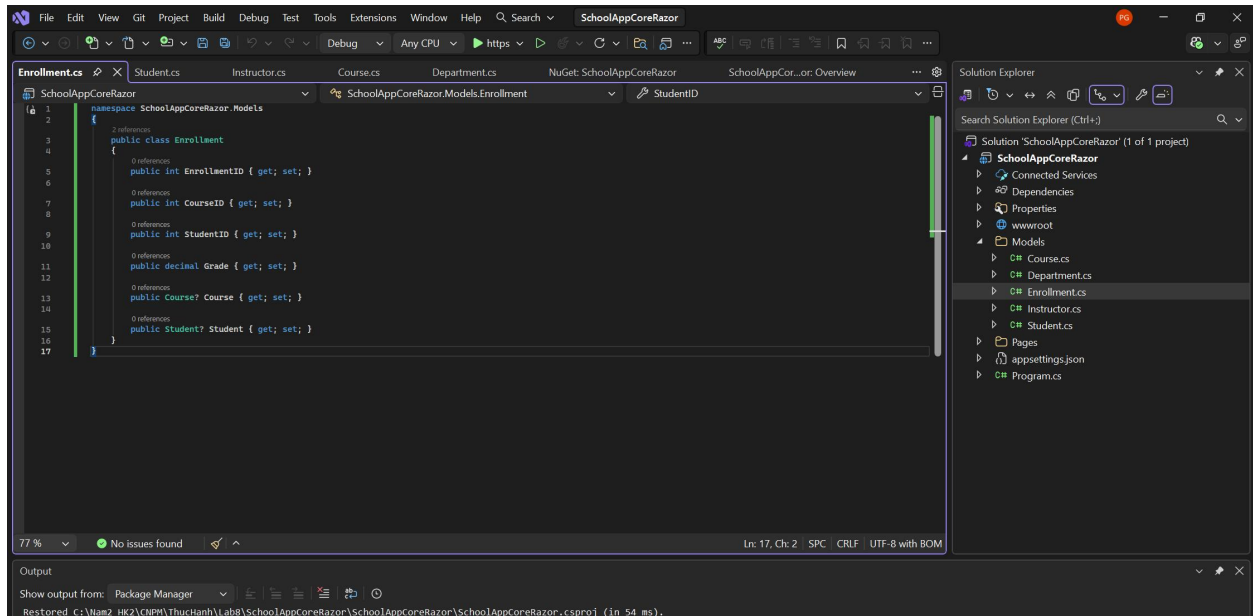


Visual Studio IDE showing the code for Department.cs:

```
1 using System;
2 using System.Collections.Generic;
3
4 namespace SchoolAppCoreRazor.Models
5 {
6     public class Department
7     {
8         public int DepartmentID { get; set; }
9
10        public required string DepartmentName { get; set; }
11
12        public decimal Budget { get; set; }
13
14        public DateTime StartDate { get; set; }
15
16        public ICollection<Course> Courses { get; set; }
17    }
18 }
```

77 % No issues found Ln: 18, Ch: 2 SPC CRLF UTF-8 with BOM





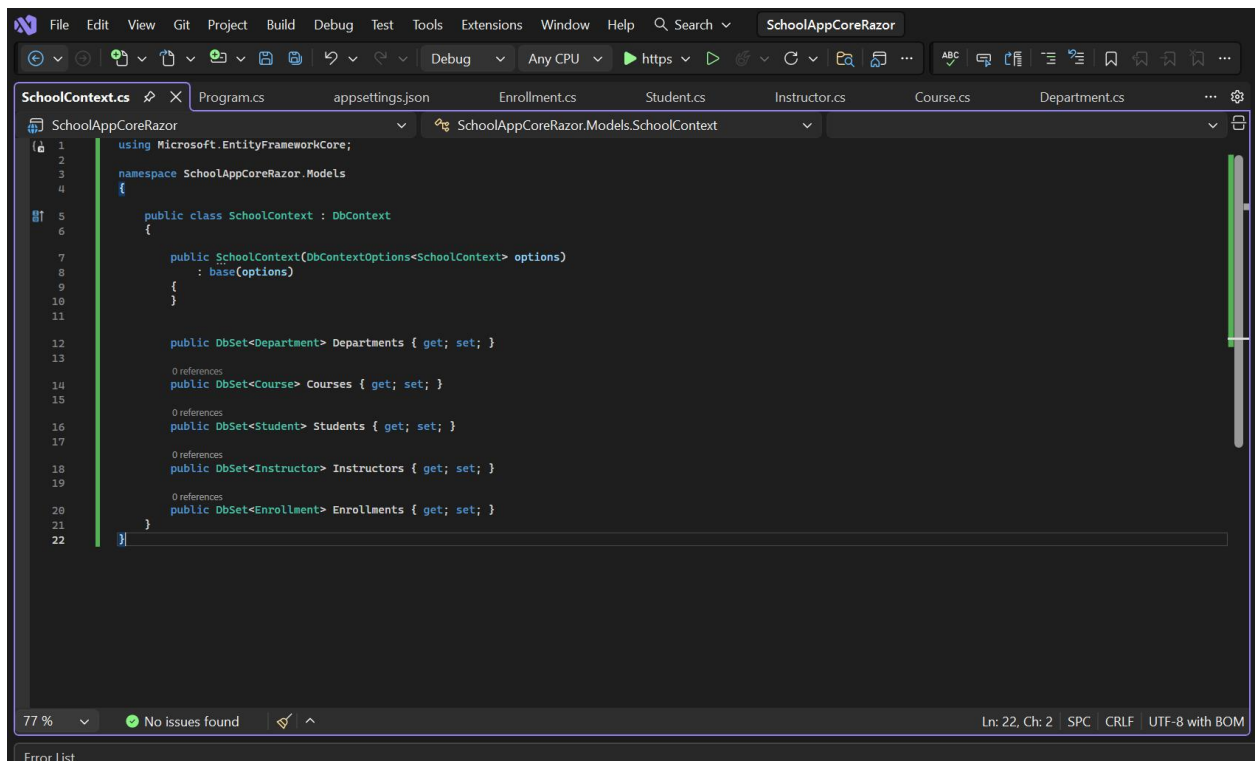
BƯỚC 4 — Tạo DbContext

Nội dung

Em tạo class SchoolContext.cs kế thừa từ DbContext.

Class này dùng để:

- kết nối database,
- quản lý các bảng dữ liệu thông qua DbSet.

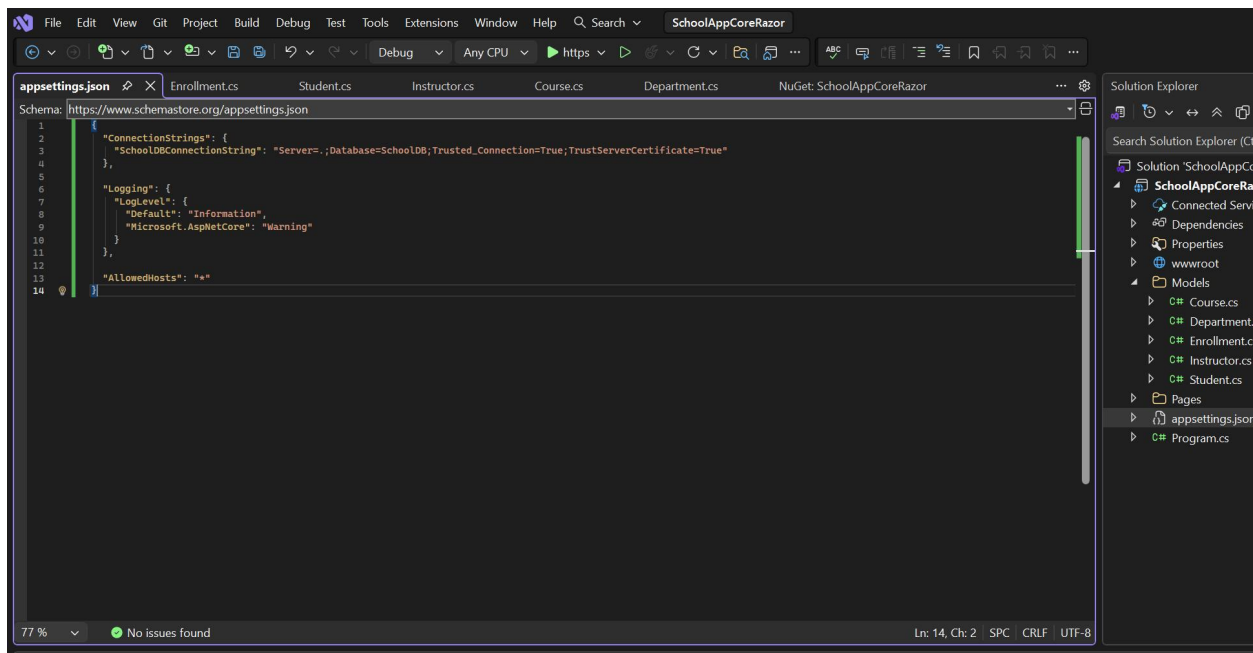


```
1 using Microsoft.EntityFrameworkCore;
2
3 namespace SchoolAppCoreRazor.Models
4 {
5     public class SchoolContext : DbContext
6     {
7         public SchoolContext(DbContextOptions<SchoolContext> options)
8             : base(options)
9         {
10         }
11
12         public DbSet<Department> Departments { get; set; }
13
14         0 references
15         public DbSet<Course> Courses { get; set; }
16
17         0 references
18         public DbSet<Student> Students { get; set; }
19
20         0 references
21         public DbSet<Instructor> Instructors { get; set; }
22
23         0 references
24         public DbSet<Enrollment> Enrollments { get; set; }
25     }
26 }
```

BƯỚC 5 — Cấu hình appsettings.json

Nội dung

Trong file appsettings.json, em cấu hình connection string để kết nối với SQL Server.

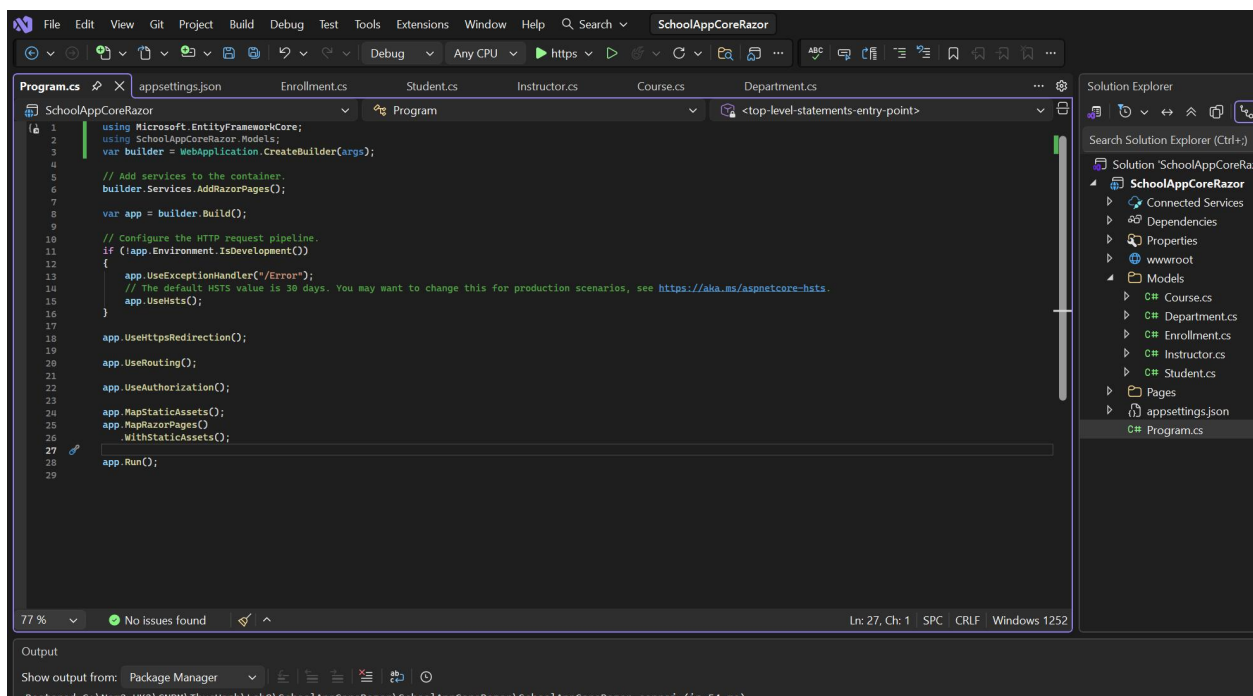


BƯỚC 6 — Cấu hình Program.cs

Nội dung

Trong file Program.cs, em thêm dịch vụ DbContext bằng Entity Framework Core.

Hệ thống sẽ sử dụng SQL Server thông qua connection string đã cấu hình.



BƯỚC 7 — Migration Database

Nội dung

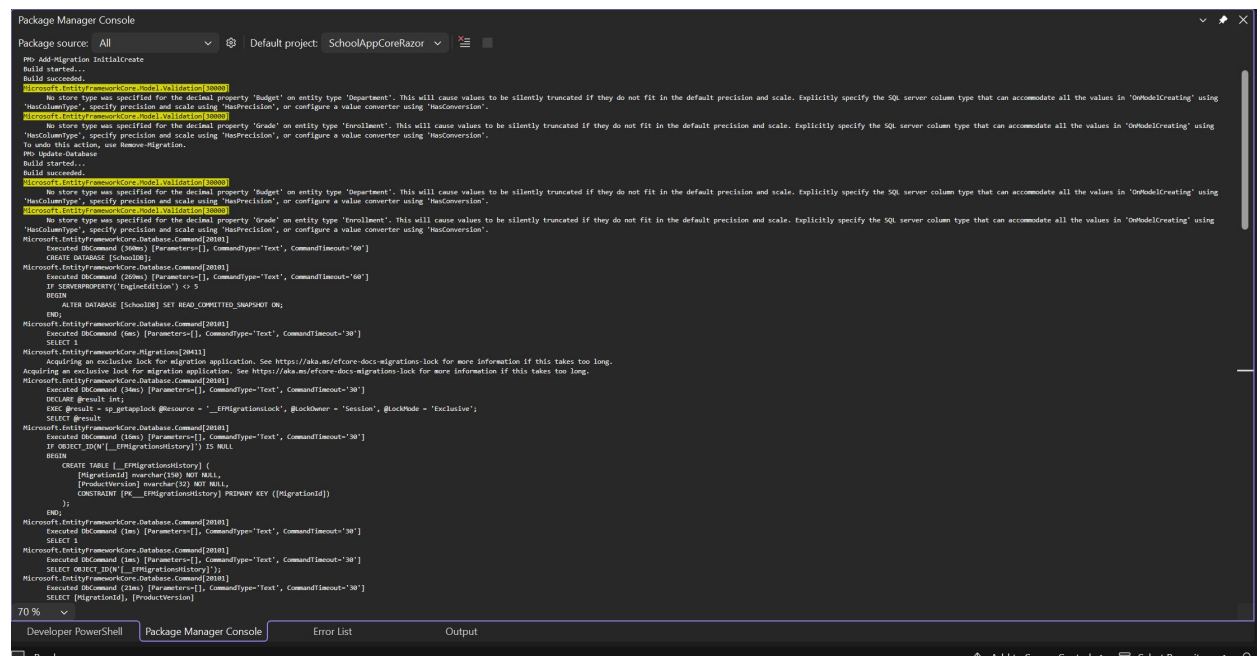
Sau khi hoàn thành model và DbContext, em sử dụng Migration để tạo database tự động.

Các lệnh sử dụng:

Add-Migration InitialCreate

Update-Database

Sau khi chạy thành công, database SchoolDB cùng các table được tạo trong SQL Server.



```
Package Manager Console
Package source: All Default project: SchoolAppCoreRazor
PM> Add-Migration InitialCreate
Build started...
Build succeeded.
Microsoft.EntityFrameworkCore.Model.Validation [30000]
No store type was specified for the decimal property 'Budget' on entity type 'Department'. This will cause values to be silently truncated if they do not fit in the default precision and scale. Explicitly specify the SQL server column type that can accommodate all the values in 'OnModelCreating' using 'HasColumnType', specify precision and scale using 'HasPrecision', or configure a value converter using 'HasConversion'.
Microsoft.EntityFrameworkCore.Model.Validation [30000]
No store type was specified for the decimal property 'Grade' on entity type 'Enrollment'. This will cause values to be silently truncated if they do not fit in the default precision and scale. Explicitly specify the SQL server column type that can accommodate all the values in 'OnModelCreating' using 'HasColumnType', specify precision and scale using 'HasPrecision', or configure a value converter using 'HasConversion'.
To undo this action, use Remove-Migration.
PM> Update-Database
Build started...
Build succeeded.
Microsoft.EntityFrameworkCore.Model.Validation [30000]
No store type was specified for the decimal property 'Budget' on entity type 'Department'. This will cause values to be silently truncated if they do not fit in the default precision and scale. Explicitly specify the SQL server column type that can accommodate all the values in 'OnModelCreating' using 'HasColumnType', specify precision and scale using 'HasPrecision', or configure a value converter using 'HasConversion'.
Microsoft.EntityFrameworkCore.Model.Validation [30000]
No store type was specified for the decimal property 'Grade' on entity type 'Enrollment'. This will cause values to be silently truncated if they do not fit in the default precision and scale. Explicitly specify the SQL server column type that can accommodate all the values in 'OnModelCreating' using 'HasColumnType', specify precision and scale using 'HasPrecision', or configure a value converter using 'HasConversion'.
Microsoft.EntityFrameworkCore.Database.Command [20101]
Executed DbCommand (10ms) [Parameters=[], CommandType='Text', CommandTimeout='60']
CREATE DATABASE [SchoolDB];
Microsoft.EntityFrameworkCore.Database.Command [20101]
Executed DbCommand (20ms) [Parameters=[], CommandType='Text', CommandTimeout='60']
IF SERVERPROPERTY('EngineEdition') <= 5
BEGIN
    ALTER DATABASE [SchoolDB] SET READ_COMMITTED_SNAPSHOT ON;
END;
Microsoft.EntityFrameworkCore.Database.Command [20101]
Executed DbCommand (0ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT 1
Microsoft.EntityFrameworkCore.Migrations [20411]
Acquiring an exclusive lock for migration application. See https://aka.ms/efcore-docs-migrations-lock for more information if this takes too long.
Acquiring an exclusive lock for migration application. See https://aka.ms/efcore-docs-migrations-lock for more information if this takes too long.
Microsoft.EntityFrameworkCore.Database.Command [20101]
Executed DbCommand (3ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
DECLARE @result int;
EXEC @result = sp_getapplock @resource = 'EFMigrationsLock', @lockOwner = 'Session', @lockMode = 'Exclusive';
SELECT @result;
Microsoft.EntityFrameworkCore.Database.Command [20101]
Executed DbCommand (10ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
IF OBJECT_ID('[_EFMigrationsHistory]') IS NULL
BEGIN
    CREATE TABLE [_EFMigrationsHistory] (
        [MigrationId] nvarchar(128) NOT NULL,
        [ProductVersion] nvarchar(32) NOT NULL,
        CONSTRAINT [PK__EFMigrationsHistory] PRIMARY KEY ([MigrationId])
    );
END;
Microsoft.EntityFrameworkCore.Database.Command [20101]
Executed DbCommand (1ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT 1
Microsoft.EntityFrameworkCore.Database.Command [20101]
Executed DbCommand (1ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT OBJECT_ID('[_EFMigrationsHistory]')
Microsoft.EntityFrameworkCore.Database.Command [20101]
Executed DbCommand (1ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT [MigrationId], [ProductVersion]
```

BƯỚC 8 — Tạo CRUD bằng Scaffold

Nội dung

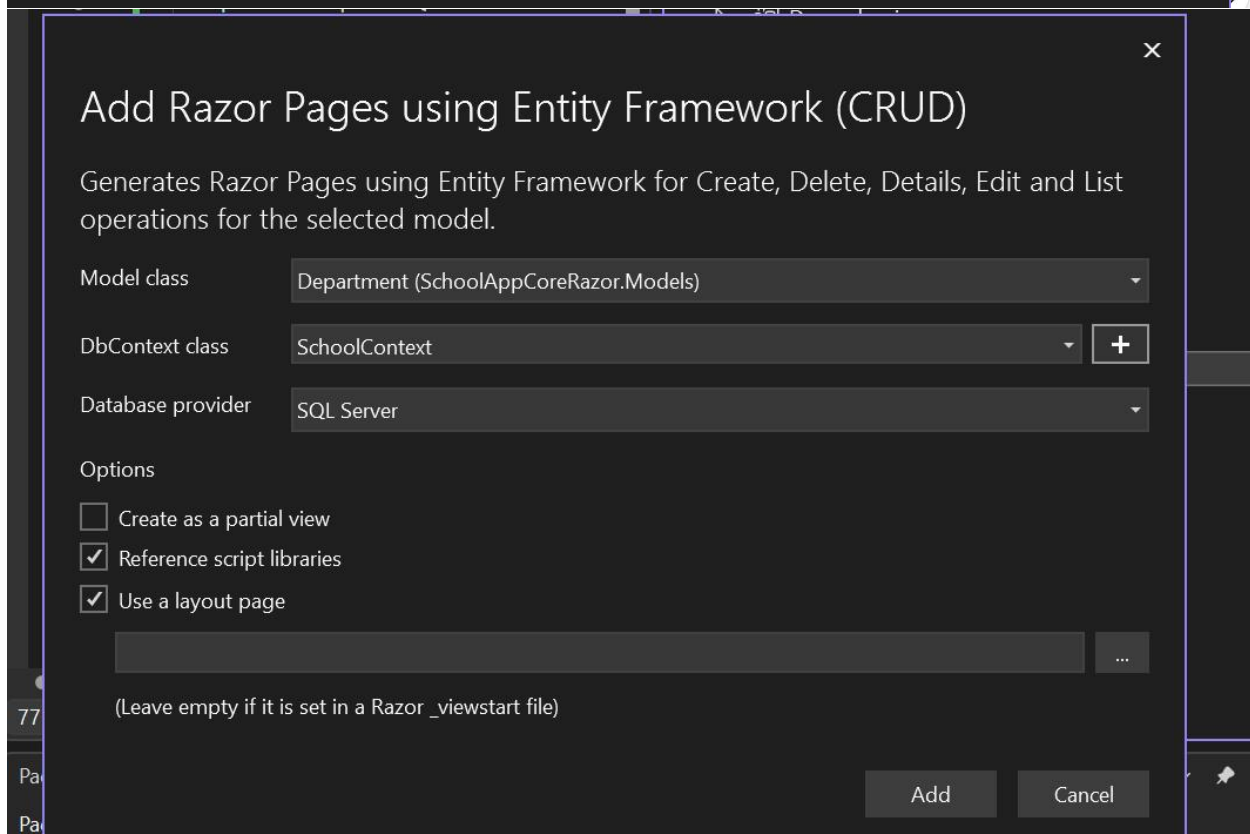
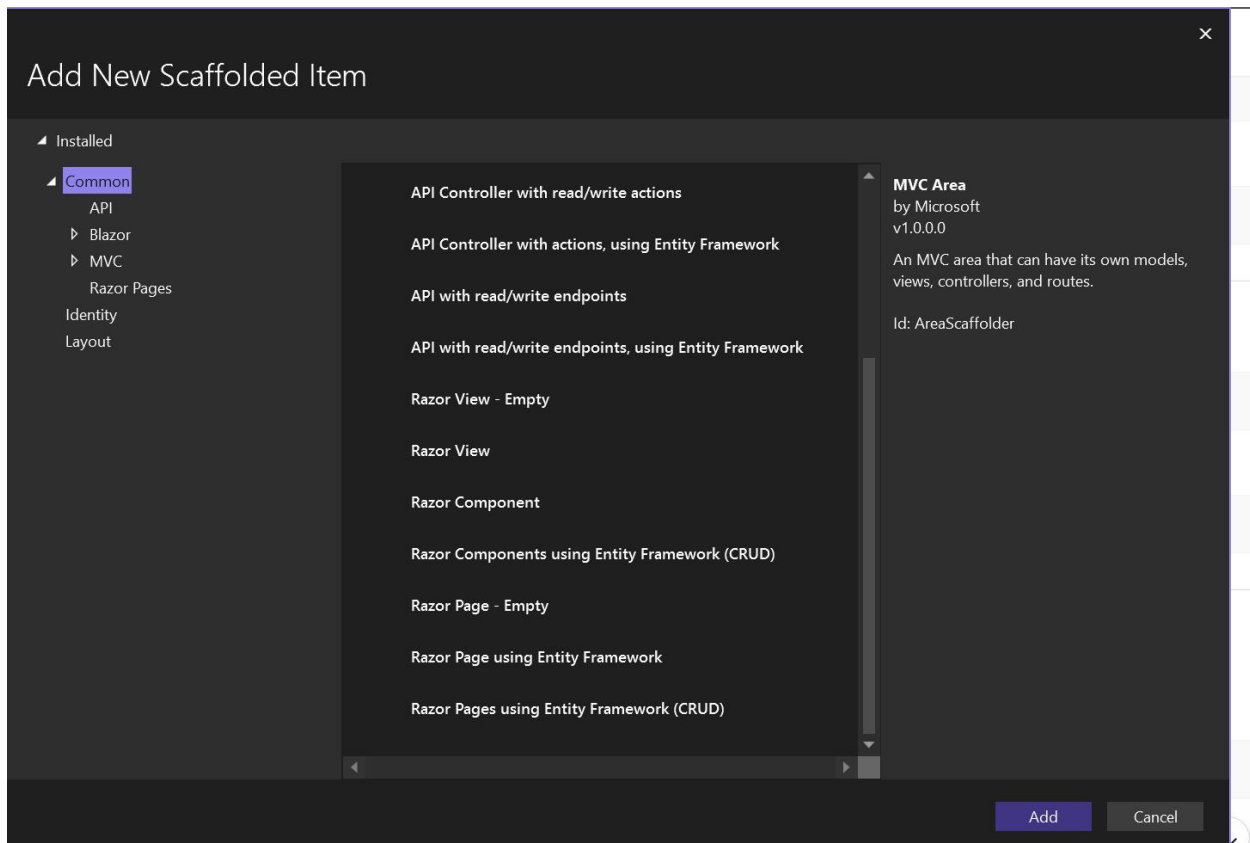
Tiếp theo, em sử dụng Scaffold để tạo CRUD tự động cho các model:

- Departments
- Students

- Courses
- Instructors
- Enrollments

Scaffold giúp tự động tạo:

- Razor Pages
- Form Create/Edit/Delete
- Code xử lý dữ liệu



BƯỚC 9 — Chỉnh sửa Layout

Nội dung

Em chỉnh sửa file `_Layout.cshtml` để thêm menu điều hướng cho hệ thống:

- Departments
- Students
- Courses
- Instructors
- Enrollments

Điều này giúp người dùng dễ dàng truy cập các chức năng CRUD.

C# SchoolAppCoreRazor

```
1 <!-- DOCTYPE html -->
2 <html lang="en" -->
3 <head>
4     <meta charset="utf-8" />
5     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6     <title>@ ViewData["Title"] - SchoolAppCoreRazor</title>
7
8     <script type="text/javascript"></script>
9
10    <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
11    <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
12    <link rel="stylesheet" href="~/SchoolAppCoreRazor.styles.css" asp-append-version="true" />
13 </head>
14
15 <body>
16
17     <div class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white border-bottom box-shadow mb-3">
18
19         <div class="container">
20
21             <div class="navbar-brand" asp-area="" asp-page="/Index">
22                 SchoolAppCoreRazor
23             </div>
24
25             <button class="navbar-toggler"
26                 type="button"
27                 data-bs-toggle="collapse"
28                 data-bs-target="#navbar-collapse"
29                 aria-controls="navbar-collapse"
30                 aria-expanded="false"
31                 aria-label="Toggle navigation">
32
33                 <span class="navbar-toggler-icon"></span>
34             </button>
35
36             <div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
37
38                 <ul class="navbar-nav flex-grow-1">
39
40                     <li class="nav-item">
41                         <a class="nav-link text-dark"
42                             asp-area=""
43                             asp-page="/Index">
44                             Home
45                         </a>
46                     </li>
47
48                     <li class="nav-item">
49                         <a class="nav-link text-dark"
50                             asp-area=""
51                             asp-page="/Privacy">
52                             Privacy
53                         </a>
54                     </li>
55
56                     <li class="nav-item">
57                         <a class="nav-link text-dark"
58                             asp-area=""
59                             asp-page="/Departments/Index">
60                             Departments
61                         </a>
62                     </li>
63
64                     <li class="nav-item">
65                         <a class="nav-link text-dark"
66                             asp-area=""
67                             asp-page="/Students/Index">
68                             Students
69                         </a>
70                     </li>
71
72                     <li class="nav-item">
73                         <a class="nav-link text-dark"
74                             asp-area=""
75                             asp-page="/Courses/Index">
76                             Courses
77                         </a>
78                     </li>
79
80                     <li class="nav-item">
81                         <a class="nav-link text-dark"
82                             asp-area=""
83                             asp-page="/Instructors/Index">
84                             Instructors
85                         </a>
86                     </li>
87
88                     <li class="nav-item">
89                         <a class="nav-link text-dark"
90                             asp-area=""
91                             asp-page="/Enrollments/Index">
92                             Enrollments
93                         </a>
94                     </li>
95
96                 </ul>
97
98             </div>
99         </div>
100
101     </div>
102
103     <div class="container">
104
105         <div role="main" class="pt-3">
106             @RenderBody()
107         </div>
108
109     </div>
110
111     <div class="border-top footer text-muted">
112
113         <div class="container">
114
115             <script> 2026 - SchoolAppCoreRazor -
116                 <a asp-area="" asp-page="/Privacy">Privacy</a>
117             </div>
118
119         </div>
120
121         <script src="~/lib/jquery/dist/jquery.min.js"></script>
122
123         <script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
124
125         <script src="~/js/site.js" asp-append-version="true"></script>
126
127         @await RenderSectionAsync("Scripts", required: false)
128     </div>
129 </body>
130 </html>
```

BƯỚC 10 — Chạy chương trình

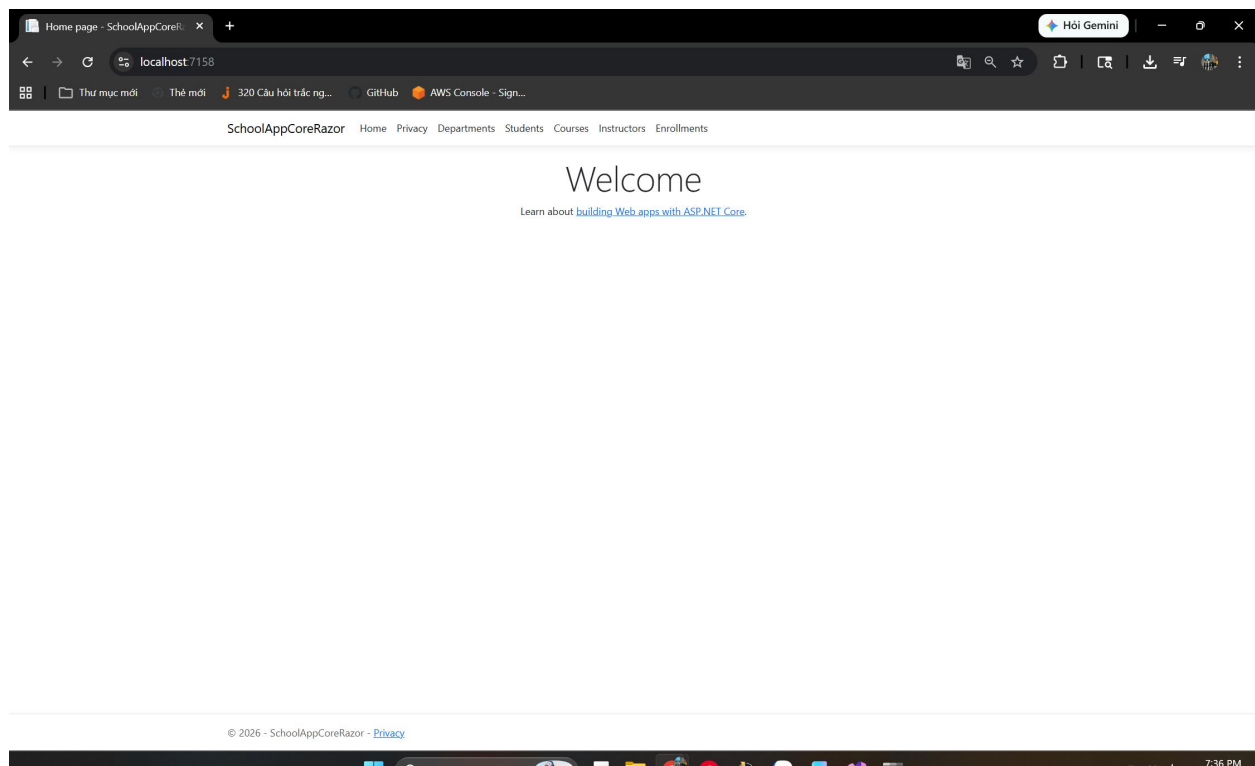
Nội dung

Sau khi hoàn thành các bước trên, em chạy chương trình bằng F5.

Hệ thống hoạt động thành công với đầy đủ chức năng:

- Create
- Read
- Update
- Delete

Dữ liệu được lưu trực tiếp vào SQL Server.

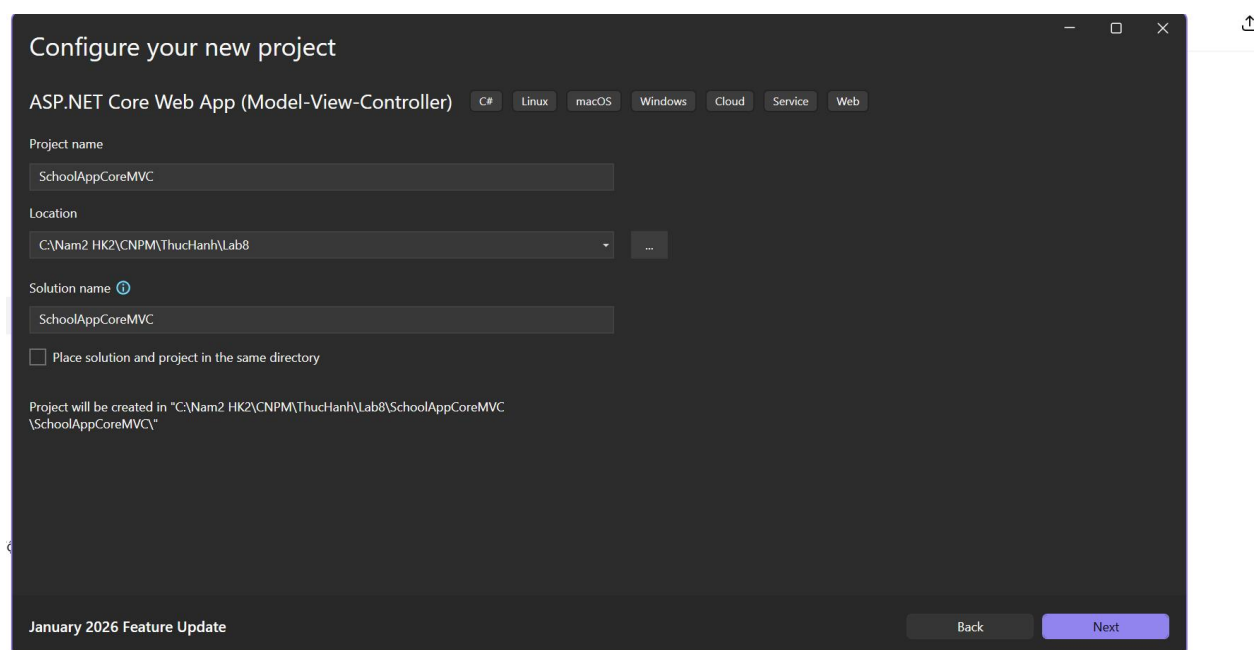


EXERCISE 2 — ASP.NET CORE MVC

BƯỚC 1 — Tạo Project MVC

Nội dung

Đầu tiên, em tạo project ASP.NET Core MVC trong Visual Studio với tên:



BƯỚC 2 — Cài đặt Entity Framework Core

Nội dung

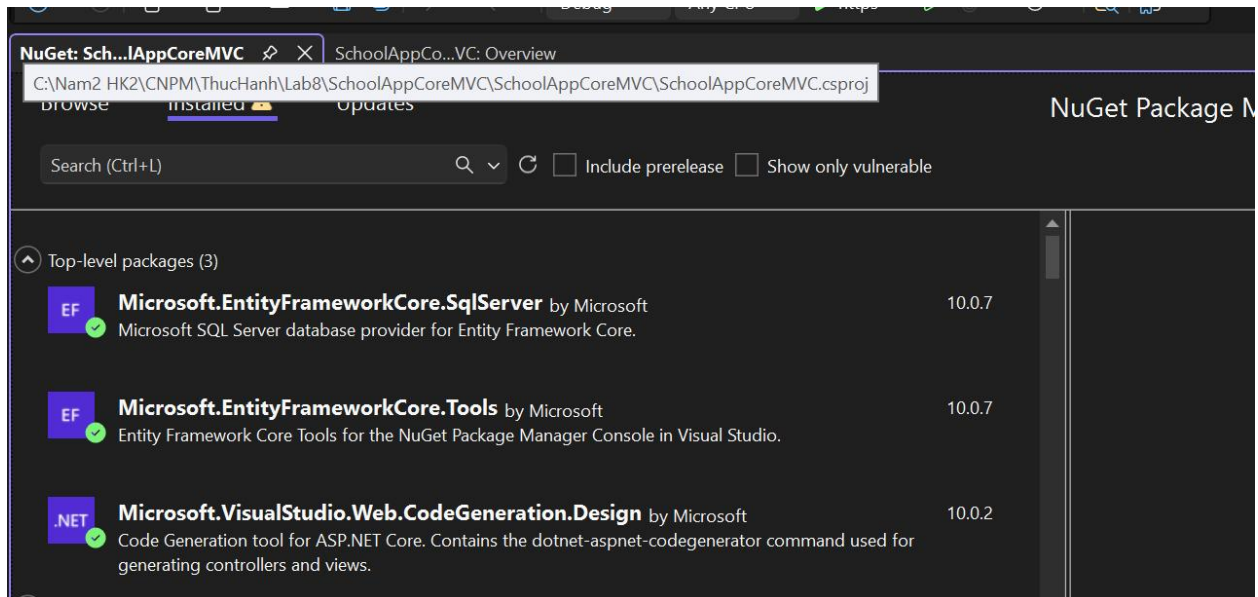
Sau khi tạo project, em cài đặt các package cần thiết bằng NuGet Package Manager:

- Microsoft.EntityFrameworkCore.SqlServer
- Microsoft.EntityFrameworkCore.Tools
- Microsoft.VisualStudio.Web.CodeGeneration.Design

Các package này hỗ trợ:

- kết nối SQL Server,

- migration database,
- scaffold CRUD tự động.



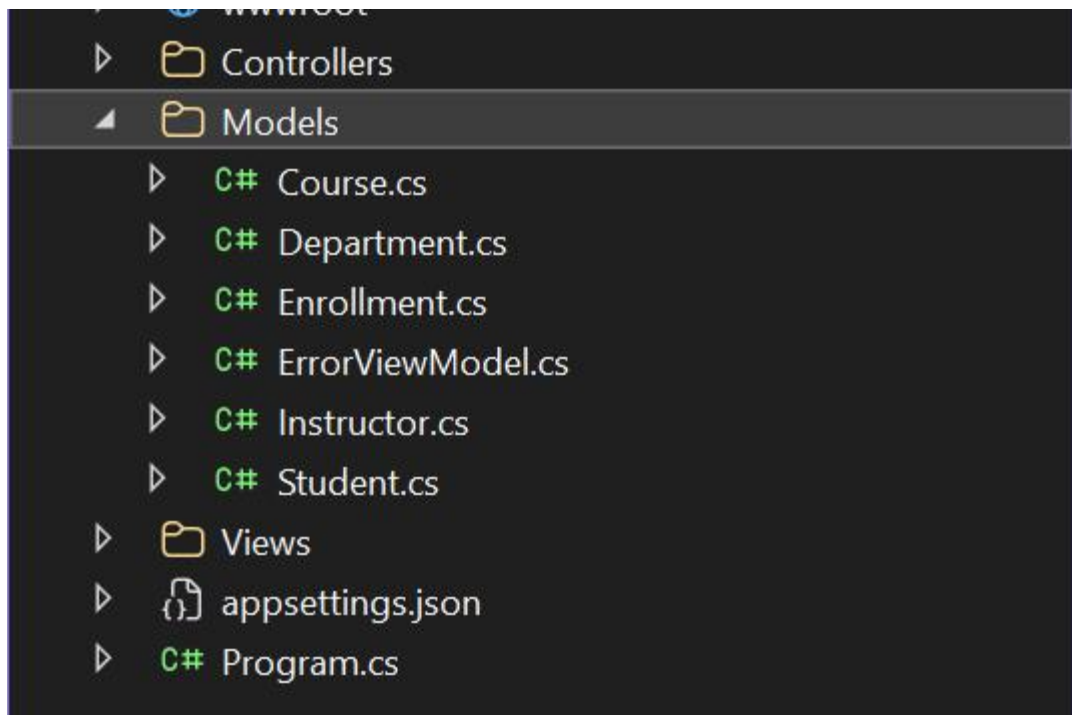
BƯỚC 3 — Tạo Models

Nội dung

Em tạo folder Models và thêm các class:

- Department.cs
- Course.cs
- Student.cs
- Instructor.cs
- Enrollment.cs

Các class dùng để mô tả dữ liệu cho hệ thống quản lý trường học.

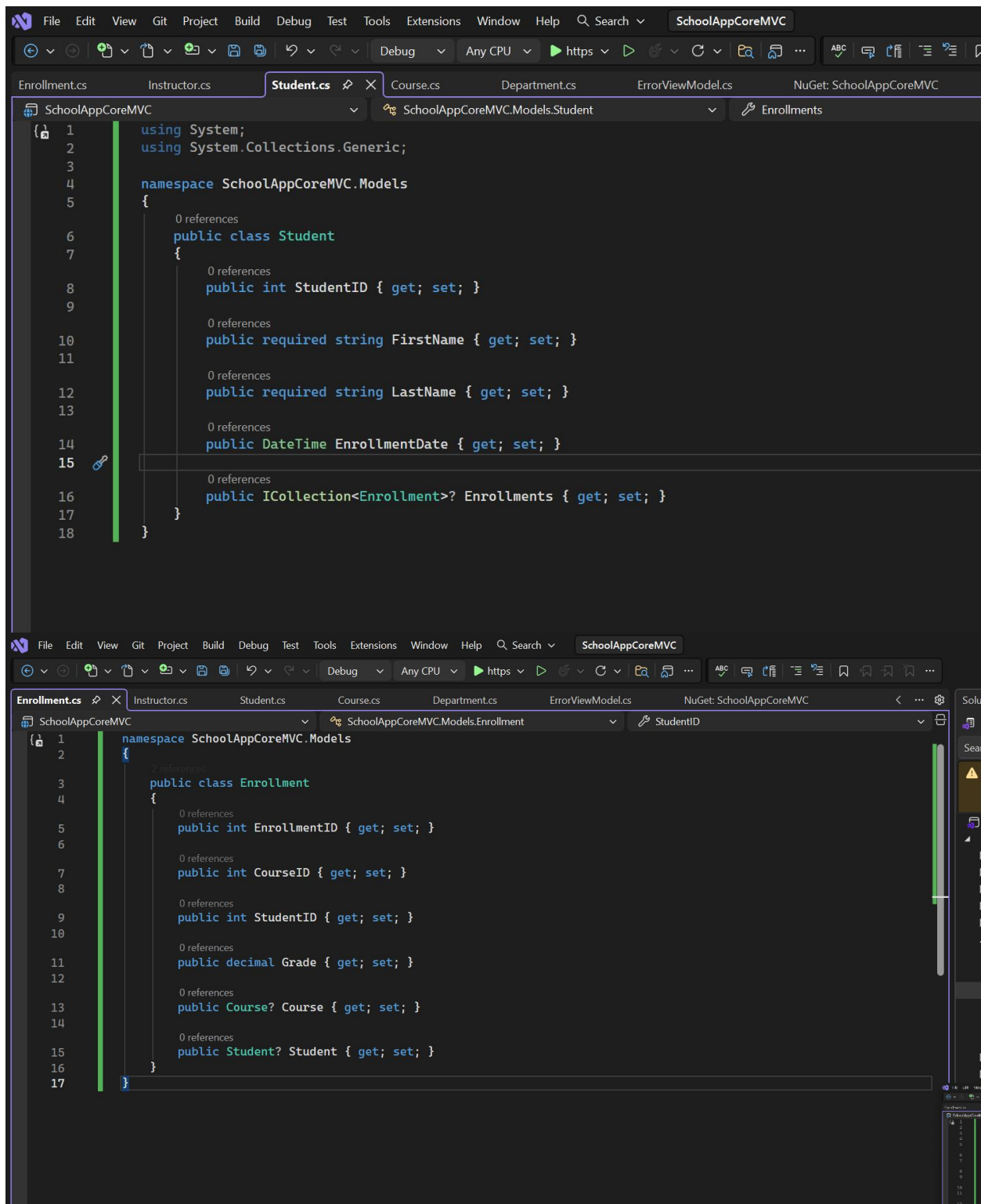


```
SchoolAppCoreMVC
SchoolAppCoreMVC.Models.Department
StartDate

1  using System;
2  using System.Collections.Generic;
3
4  namespace SchoolAppCoreMVC.Models
5  {
6      0 references
7      public class Department
8      {
9          0 references
10         public int DepartmentID { get; set; }
11
12         0 references
13         public required string DepartmentName { get; set; }
14
15         0 references
16         public decimal Budget { get; set; }
17
18         0 references
19         public DateTime StartDate { get; set; }
20
21         0 references
22         public ICollection<Course>? Courses { get; set; }
23     }
24 }
```

```
Enrollment.cs | Instructor.cs | Student.cs | Course.cs | Department.cs | ErrorViewModel.cs | NuGet: SchoolAppCoreMVC
SchoolAppCoreMVC | SchoolAppCoreMVC.Models.Course
1  using System.Collections.Generic;
2
3  namespace SchoolAppCoreMVC.Models
4  {
5      public class Course
6      {
7          public int CourseID { get; set; }
8
9          public required string Title { get; set; }
10
11         public int Credits { get; set; }
12
13         public int DepartmentID { get; set; }
14
15         0 references
16         public Department? Department { get; set; }
17
18         0 references
19         public ICollection<Enrollment>? Enrollments { get; set; }
20     }
21 }

Enrollment.cs | Instructor.cs | Student.cs | Course.cs | Department.cs | ErrorViewModel.cs | NuGet: SchoolAppCoreMVC
SchoolAppCoreMVC | SchoolAppCoreMVC.Models.Instructor | LastName
1  using System;
2
3  namespace SchoolAppCoreMVC.Models
4  {
5      public class Instructor
6      {
7          0 references
8          public int InstructorID { get; set; }
9
10         0 references
11         public required string FirstName { get; set; }
12
13         0 references
14         public required string LastName { get; set; }
15
16         0 references
17         public DateTime HireDate { get; set; }
18     }
19 }
```



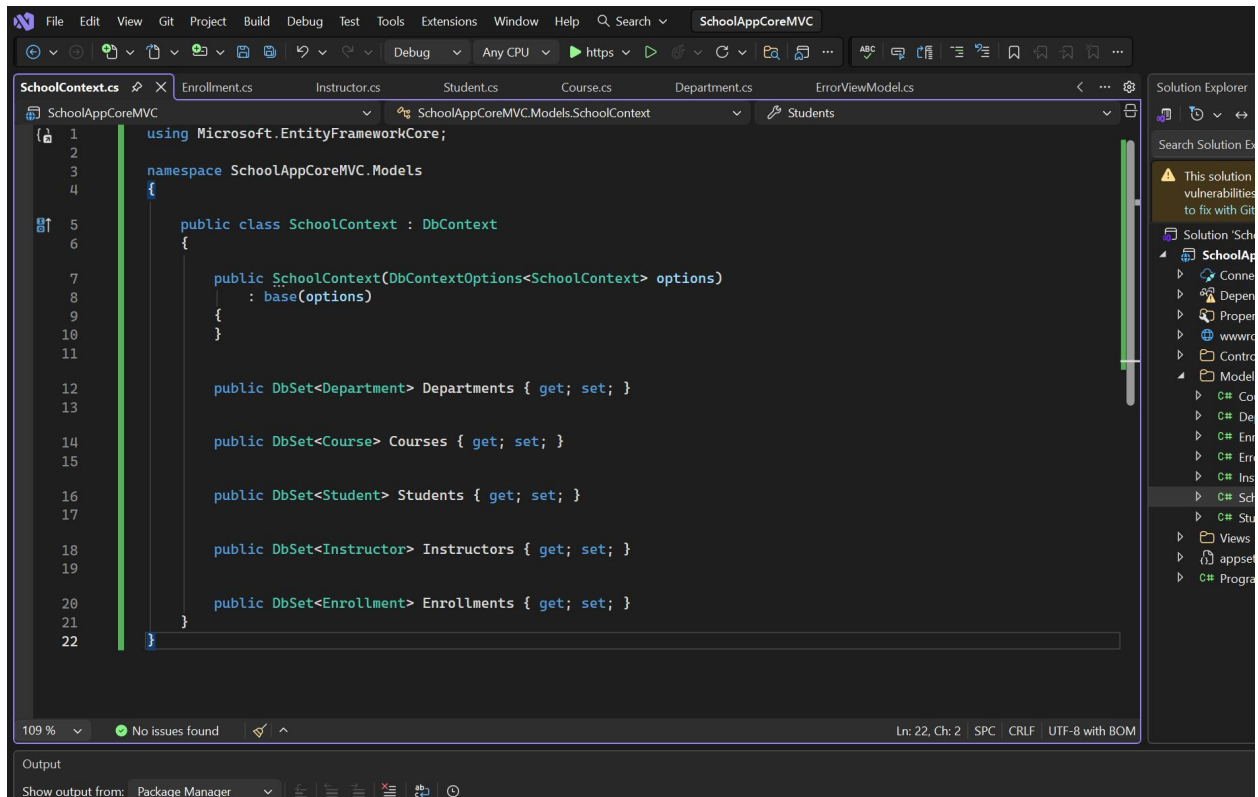
BƯỚC 4 — Tạo DbContext

Nội dung

Em tạo class SchoolContext.cs kế thừa từ DbContext.

Class này giúp:

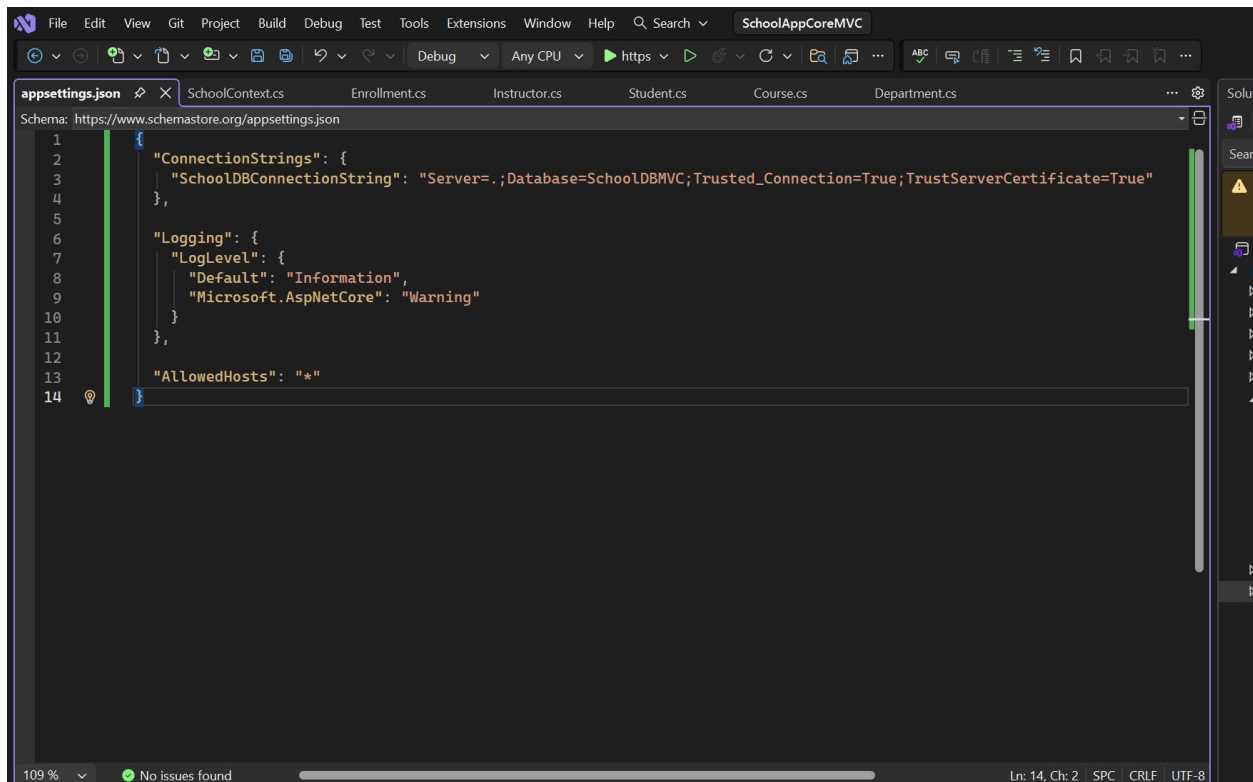
- quản lý kết nối database,
- tạo các bảng dữ liệu thông qua DbSet.



BƯỚC 5 — Cấu hình appsettings.json

Nội dung

Trong file appsettings.json, em cấu hình connection string để kết nối SQL Server.



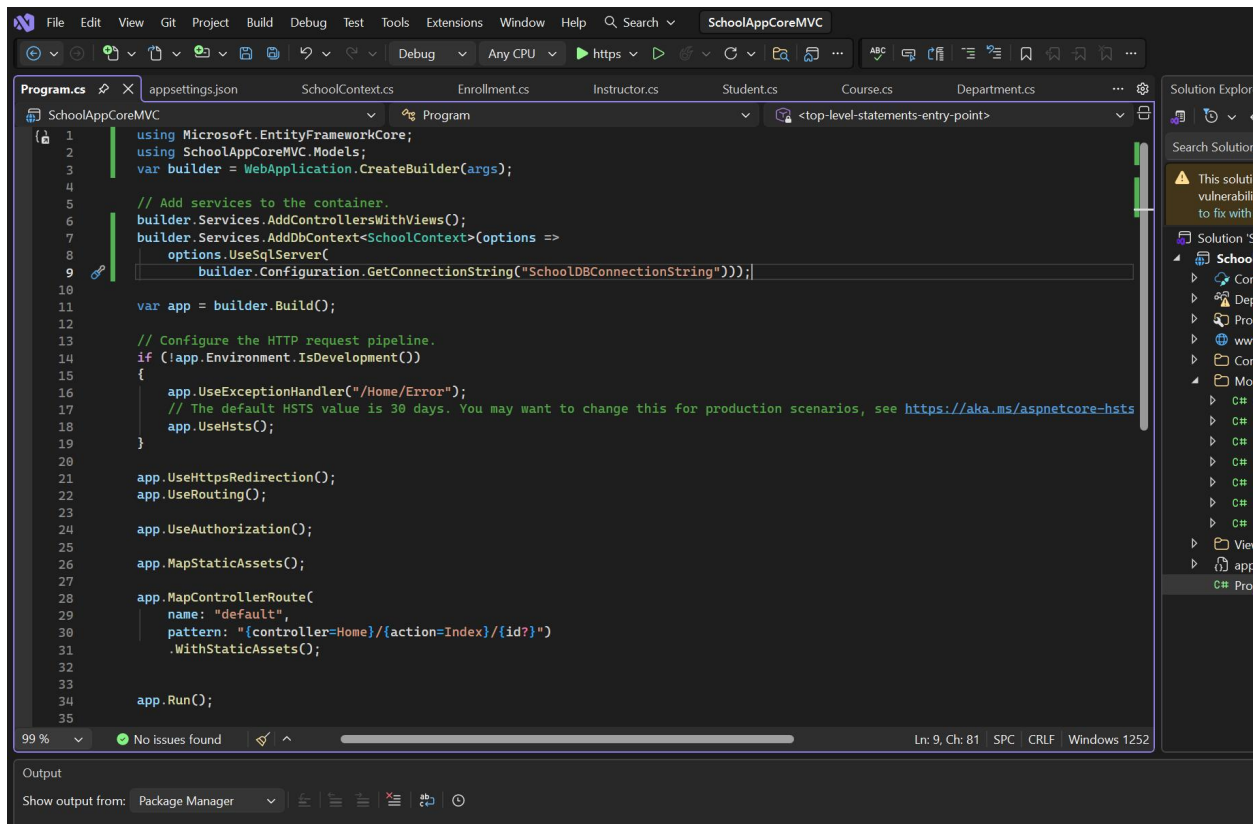
```
1 {
2   "ConnectionStrings": {
3     "SchoolDBConnectionString": "Server=.;Database=SchoolDBMVC;Trusted_Connection=True;TrustServerCertificate=True"
4   },
5
6   "Logging": {
7     "LogLevel": {
8       "Default": "Information",
9       "Microsoft.AspNetCore": "Warning"
10    }
11  },
12
13  "AllowedHosts": "*"
14 }
```

BƯỚC 6 — Cấu hình Program.cs

Nội dung

Trong file Program.cs, em thêm DbContext vào service container bằng Entity Framework Core.

Ứng dụng sử dụng SQL Server thông qua connection string đã cấu hình.



BƯỚC 7 — Migration Database

Nội dung

Sau khi hoàn thành model và DbContext, em sử dụng migration để tạo database tự động.

Các lệnh sử dụng:

Add-Migration InitialCreate

Update-Database

Sau khi thực hiện thành công, database SchoolDBMVC cùng các bảng dữ liệu được tạo trong SQL Server.

Package Manager Console

Package source: All Default project: SchoolAppCoreMVC

PM> Add-Migration InitialCreate
Build started...
Build succeeded.
Microsoft.EntityFrameworkCore.Model.Validation[30000]
No store type was specified for the decimal property 'Budget' on entity type 'Department'. This will cause values to be silently truncated if they do not fit in the default precision and scale. Explicitly specify the SQL server column type that can accommodate all the values in 'OnModelCreating' using 'HasColumnType', specify precision and scale using 'HasPrecision', or configure a value converter using 'HasConversion'.
Microsoft.EntityFrameworkCore.Model.Validation[30000]
No store type was specified for the decimal property 'Grade' on entity type 'Enrollment'. This will cause values to be silently truncated if they do not fit in the default precision and scale. Explicitly specify the SQL server column type that can accommodate all the values in 'OnModelCreating' using 'HasColumnType', specify precision and scale using 'HasPrecision', or configure a value converter using 'HasConversion'.
To undo this action, use Remove-Migration.
PM> Update-Database
Build started...
Build succeeded.
Microsoft.EntityFrameworkCore.Model.Validation[30000]
No store type was specified for the decimal property 'Budget' on entity type 'Department'. This will cause values to be silently truncated if they do not fit in the default precision and scale. Explicitly specify the SQL server column type that can accommodate all the values in 'OnModelCreating' using 'HasColumnType', specify precision and scale using 'HasPrecision', or configure a value converter using 'HasConversion'.
Microsoft.EntityFrameworkCore.Model.Validation[30000]
No store type was specified for the decimal property 'Grade' on entity type 'Enrollment'. This will cause values to be silently truncated if they do not fit in the default precision and scale. Explicitly specify the SQL server column type that can accommodate all the values in 'OnModelCreating' using 'HasColumnType', specify precision and scale using 'HasPrecision', or configure a value converter using 'HasConversion'.
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (256ms) [Parameters=[], CommandType='Text', CommandTimeout='60']
CREATE DATABASE [SchoolDBMVC];
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (179ms) [Parameters=[], CommandType='Text', CommandTimeout='60']
IF SERVERPROPERTY('EngineEdition') <= 5
BEGIN
ALTER DATABASE [SchoolDBMVC] SET READ_COMMITTED_SNAPSHOT ON;
END;
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (5ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT 1
Microsoft.EntityFrameworkCore.Migrations[20411]
Acquiring an exclusive lock for migration application. See https://aka.ms/efcore-docs-migrations-lock for more information if this takes too long.
Acquiring an exclusive lock for migration application. See https://aka.ms/efcore-docs-migrations-lock for more information if this takes too long.
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (24ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
DECLARE @result int;
EXEC @result = sp_getapplock @Resource = '_EFMigrationsLock', @LockOwner = 'Session', @LockMode = 'Exclusive';
SELECT @result
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (9ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
IF OBJECT_ID(N'[_EFMigrationsHistory]') IS NULL
BEGIN
CREATE TABLE [_EFMigrationsHistory] (
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (1ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT 1
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (1ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT OBJECT_ID(N'[_EFMigrationsHistory]');
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (14ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT [MigrationId], [ProductVersion]
FROM [_EFMigrationsHistory]
ORDER BY [MigrationId];
Microsoft.EntityFrameworkCore.Migrations[20402]
Applying migration '20260512125012.InitialCreate'.
Applying migration '20260512125012.InitialCreate'.
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (4ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
CREATE TABLE [Departments] (
[DepartmentID] int NOT NULL IDENTITY,
[DepartmentName] nvarchar(max) NOT NULL,
[Budget] decimal(18,2) NOT NULL,
[StartDate] datetime2 NOT NULL,
CONSTRAINT [PK_Departments] PRIMARY KEY ([DepartmentID])
);
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (5ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
CREATE TABLE [Instructors] (
[InstructorID] int NOT NULL IDENTITY,
[FirstName] nvarchar(max) NOT NULL,
[LastName] nvarchar(max) NOT NULL,
[HireDate] datetime2 NOT NULL,
CONSTRAINT [PK_Instructors] PRIMARY KEY ([InstructorID])

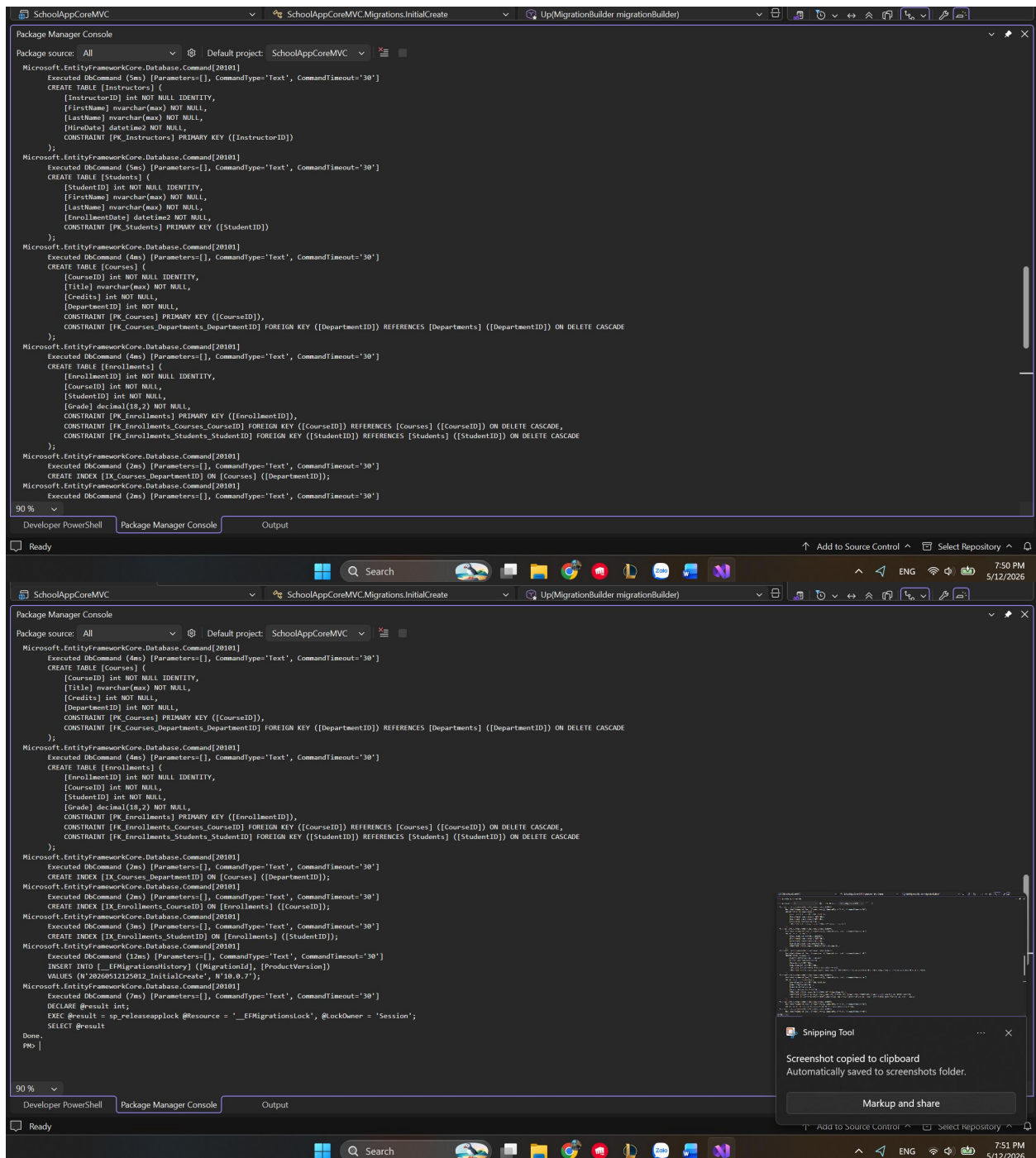
Developer PowerShell Package Manager Console Output

Package source: All Default project: SchoolAppCoreMVC

DECLARE @result int;
EXEC @result = sp_getapplock @Resource = '_EFMigrationsLock', @LockOwner = 'Session', @LockMode = 'Exclusive';
SELECT @result
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (9ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
IF OBJECT_ID(N'[_EFMigrationsHistory]') IS NULL
BEGIN
CREATE TABLE [_EFMigrationsHistory] (
[MigrationId] nvarchar(150) NOT NULL,
[ProductVersion] nvarchar(32) NOT NULL,
CONSTRAINT [PK__EFMigrationsHistory] PRIMARY KEY ([MigrationId])
);
END;
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (1ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT 1
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (1ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT OBJECT_ID(N'[_EFMigrationsHistory]');
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (14ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT [MigrationId], [ProductVersion]
FROM [_EFMigrationsHistory]
ORDER BY [MigrationId];
Microsoft.EntityFrameworkCore.Migrations[20402]
Applying migration '20260512125012.InitialCreate'.
Applying migration '20260512125012.InitialCreate'.
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (4ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
CREATE TABLE [Departments] (
[DepartmentID] int NOT NULL IDENTITY,
[DepartmentName] nvarchar(max) NOT NULL,
[Budget] decimal(18,2) NOT NULL,
[StartDate] datetime2 NOT NULL,
CONSTRAINT [PK_Departments] PRIMARY KEY ([DepartmentID])
);
Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (5ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
CREATE TABLE [Instructors] (
[InstructorID] int NOT NULL IDENTITY,
[FirstName] nvarchar(max) NOT NULL,
[LastName] nvarchar(max) NOT NULL,
[HireDate] datetime2 NOT NULL,
CONSTRAINT [PK_Instructors] PRIMARY KEY ([InstructorID])

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.



BƯỚC 8 — Tạo CRUD bằng Scaffold

Nội dung

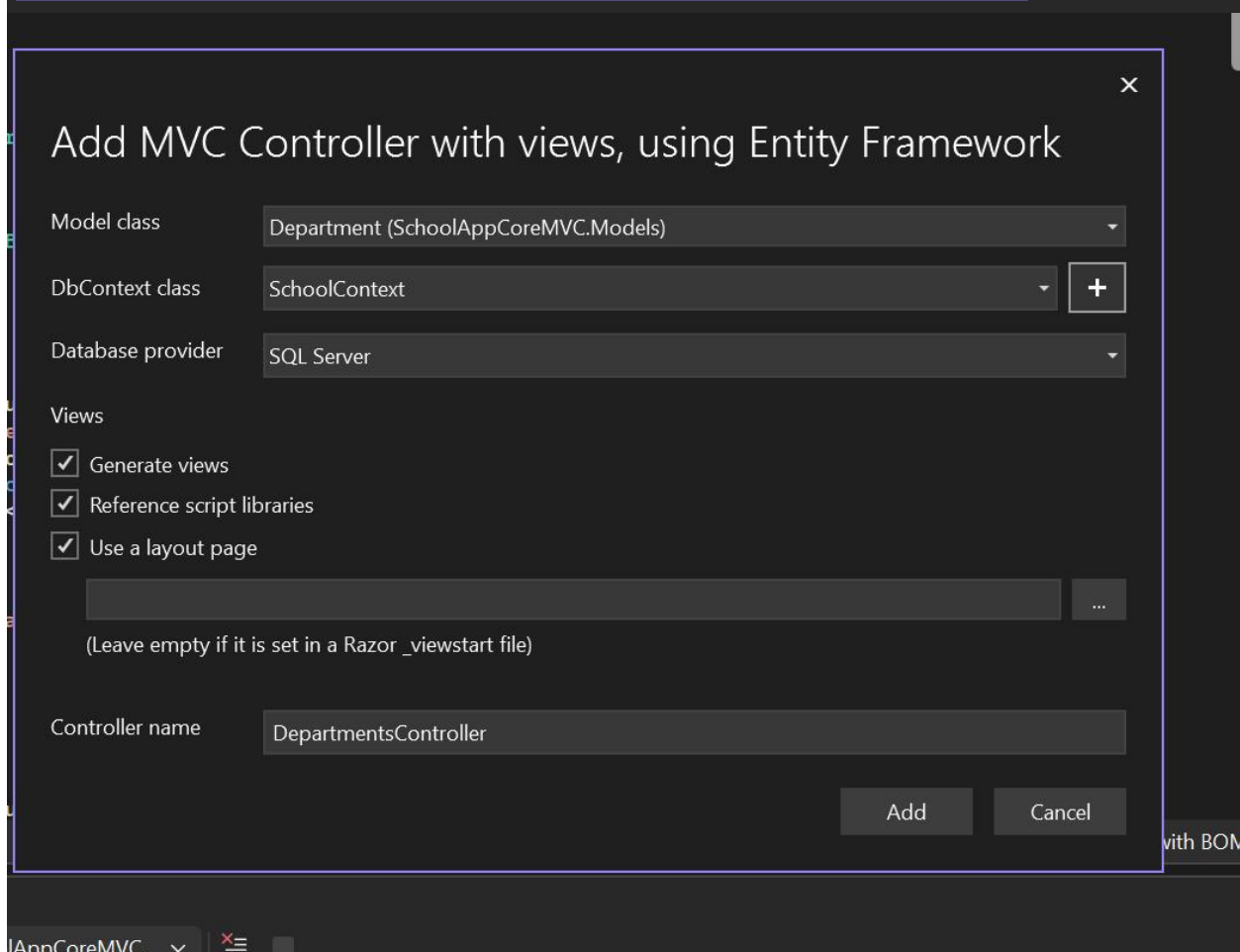
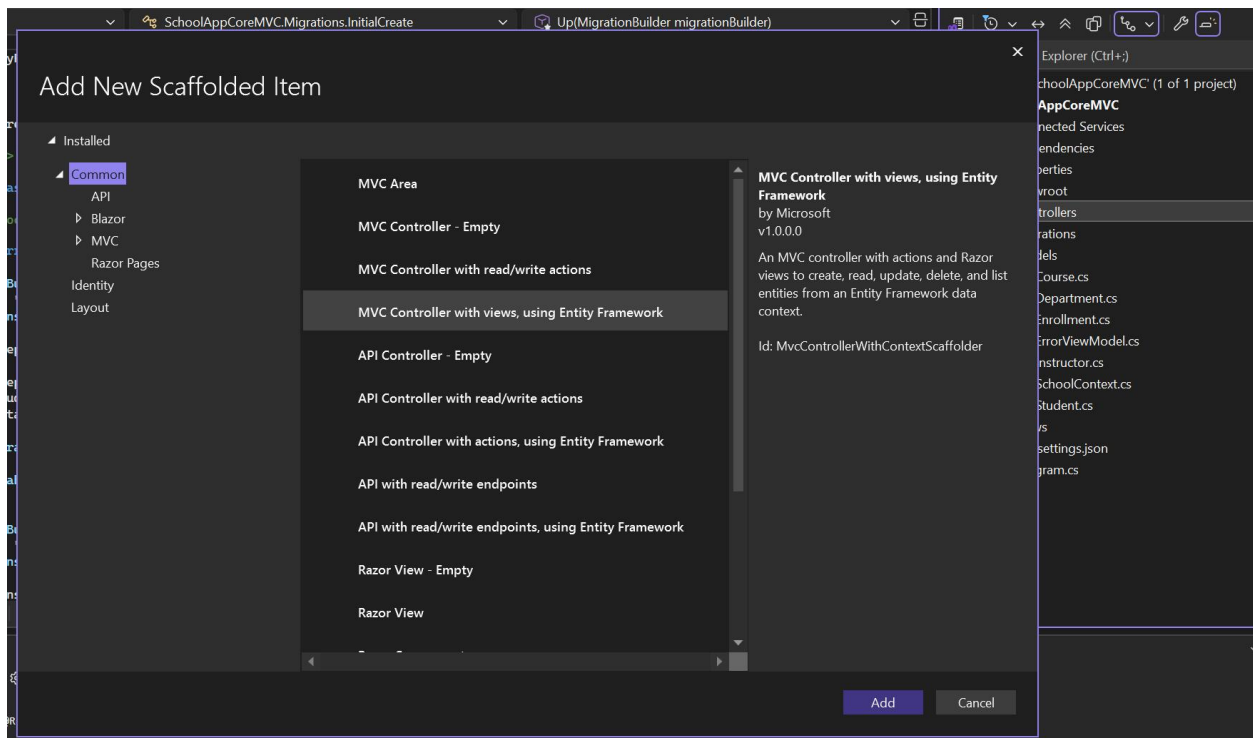
Tiếp theo, em sử dụng Scaffold để tạo CRUD tự động cho các model:

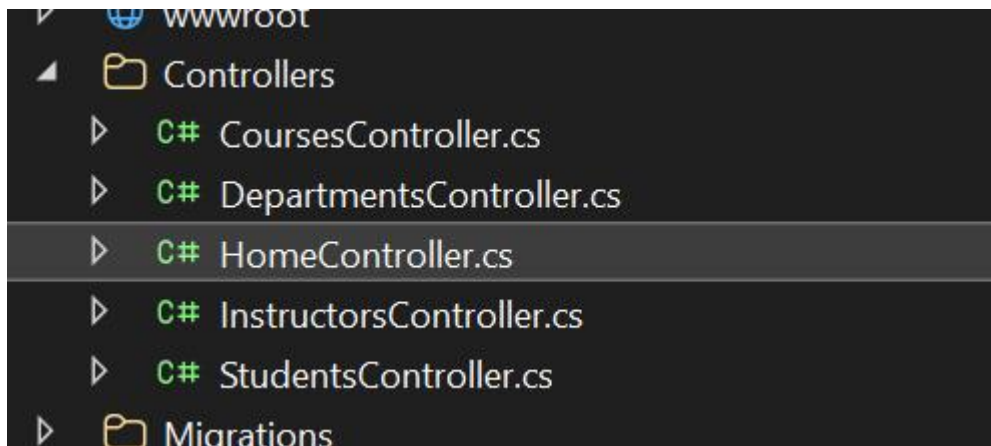
- Department

- Student
- Course
- Instructor
- Enrollment

Scaffold giúp tự động tạo:

- Controllers
- Views
- Form Create/Edit/Delete
- Code xử lý dữ liệu





BƯỚC 9 — Chỉnh sửa Layout

Nội dung

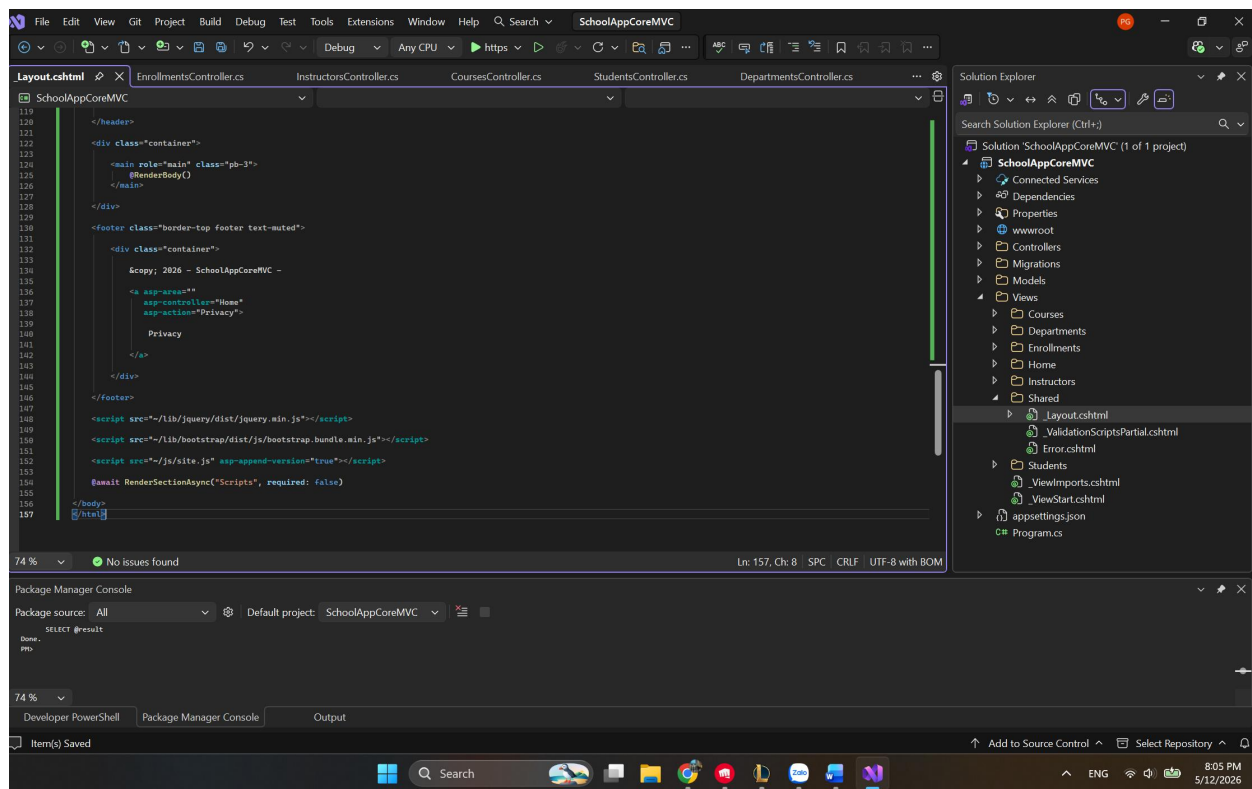
Em chỉnh sửa file:

Views/Shared/_Layout.cshtml

để thêm menu điều hướng:

- Departments
- Students
- Courses
- Instructors
- Enrollments

Điều này giúp người dùng dễ dàng truy cập các chức năng CRUD.



BƯỚC 10 — Chạy chương trình

Nội dung

Sau khi hoàn thành các bước trên, em chạy chương trình bằng F5.

Hệ thống hoạt động thành công với đầy đủ chức năng:

- Create
- Read
- Update
- Delete

Dữ liệu được lưu trực tiếp vào SQL Server.

Welcome

Learn about [building Web apps with ASP.NET Core](#).